



Standard Handbook v 2.0¹

International Soil Judging Contest

(Dates)

(Location)

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Official Handbook of the (#) International Soil Judging Contest

(Dates)

(Location)

This handbook was compiled by the (#) IUSS International Soil Judging Working Group and the host nation organizers based on the IUSS Standard Handbook v.2.0 and the necessary adaptations for the (host nation) event.

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GENERAL INFORMATION

As part of the celebrations of the (#) World Congress of Soil Science, the (#) International Soil Judging Contest has been organized for (dates) in (location). The occasion is a great opportunity for students, researchers and people interested in soils from around the world to interact and experience some of the landscapes and soils of (host nation). The scope of the Soil Judging Contest is for participants to use their knowledge and practical skills to describe, understand, classify and interpret soil characteristics in the field. There are two contests: the group contest, where the whole team acts as a group, and the individual contest. Participants (in the form of groups and individuals) will describe a series of contest profiles using basic field tools, selected standards, and guidelines. The winners will be determined based on their ability to correctly describe each soil, denominate the layers, classify the soil, evaluate potential soil functions and interpret their capacity to perform under different land use and management practices. To prepare for the event, a four-day technical training course is offered that is compulsory for the participants.

The goals of this event are to give motivated students an opportunity to assess soils in a different part of the world, to give students an opportunity to develop networks in the soil science community, and to demonstrate the career opportunities that soil science offers. This event will also encourage the wider adoption of the discipline of soil judging around the world.

The (#) ISJC will consist of four days of practice on soil and landscape description, classification and interpretation, followed by one contest day. Each team will have 1-2 accompanying academic coaches, who will assist participants in the field during the training days, but not during the contest day. The contest day will consist of description, classification and interpretation of two soil profiles and landscapes performed as a group and two soil profiles and landscapes performed individually.

Eligibility: A student must have been enrolled in a graduate program during the year of the contest or in an undergraduate program within one year prior to the contest. The teams are nominated by a national Soil Science Society. The members must be a national of the nominating country and/or a student of a university in the nominating country.

Soil Profiles: The profile pits should be at least 2 m wide and horizontally homogeneous. It must be easy to move in and out the profile pit. It is an advantage if they can be reached in short time from the accommodation. The profiles visited on one day should be close together but not so close that one can hear at one profile what is spoken at another one. An adequate place that can be used as a toilet must be available.

Training (4 days)

An international team of soil experts will give an overview of the site and profile description guidelines and soil classification standards. Local soil experts will introduce the landscape and soil conditions of (location), with a focus on the area of the contest. Short classroom sessions will be followed by practical training in field procedures and techniques. Standard samples will be provided for training in texture class and clay and sand percentage estimation. The soils in question will cover a diverse geographical area with a range of topographic, parent material and soil moisture regime conditions. The soils most commonly found in the (region name) region are (List WRB Reference Soil Groups) according to the World Reference Base of Soil Resources (WRB) and (List Soil Taxonomy Orders) according to Soil Taxonomy (ST).

The Contest (1 day)

Participants (groups and individuals) will describe and classify the contest profiles, based on the available field tools and selected standards and guidelines. Evaluation of potential soil functions and interpretation of the capacity of the soil to perform under different use and management will be also scored. The four contest soil profiles will be selected from soils similar to those seen during training.

Equipment and Reference Materials

The following equipment will be available to borrow from the pit monitor, but in a limited supply, we therefore encourage each team to bring their own.

Required for all contests:

- Soil knife/digging tool
- Magnifying glasses or hand lens
- Water bottle (extra water will be provided at each site)
- Clinometer
- Clipboard
- Munsell® Soil Color Chart
- Simple calculator (no functions other than calculating)

Optional:

- 2-mm sieve
- Hand towel

The teams are requested to bring their own Munsell® Soil Color Chart books, as only a limited number can be supplied by the organizers. The soil knife provided will be very small and simple, so we suggest the competitors to bring their own. HCl in dropper bottles and containers for soil samples (muffin bakers) will be provided by the organizers.

All the participants are encouraged to bring their own sunscreen, insect repellent, raincoat, boots, hat, and any other important personal items; none will be provided by the organizing team.

Besides this printed handbook, the Simplified WRB Diagnostics addendum, and the Simplified Keys to Soil Taxonomy addendum, the following printed reference materials will be permitted during the contest (but not provided by the organizing team):

- IUSS Working Group WRB. 2022. World Reference Base for Soil Resources, 4th edition
https://wrb.isric.org/files/WRB_fourth_edition_2022-12-18_errata_correction_2024-09-24.pdf
- Switoniak M et al. 2022. Illustrated Handbook of WRB Soil Classification
https://www.researchgate.net/publication/368839948_Illustrated_Handbook_of_WRB_Soil_Classification
- Soil Survey Staff. 2024. Field Book for Describing and Sampling Soils, Version 4.0
<https://www.nrcs.usda.gov/sites/default/files/2025-01/Field-Book-for-Describing-and-Sampling-Soils-v4.pdf>
- Soil Survey Staff. 2022: Keys to Soil Taxonomy, 13th Edition
<https://www.nrcs.usda.gov/sites/default/files/2022-09/Keys-to-Soil-Taxonomy.pdf>
- Soil Survey Staff. 2015: The Illustrated Guide to Soil Taxonomy, Version 2.0
https://www.nrcs.usda.gov/sites/default/files/2022-06/Illustrated_Guide_to_Soil_Taxonomy.pdf

Older editions of these materials are not permitted. Additional reference materials may be developed by the IUSS Working Group on International Soil Judging Contests.

Conduct of the Training

A typical conduct of training can be like this: On the 1st day there will be classroom presentations on the soils of (host nation) as well as an introduction to important soil interpretations of soils in the region. After lunch there will be a field training. On the 2nd day training on soil classification will take place focusing on the diagnostic elements of the World Reference Base for Soil Resources (WRB) and Soil Taxonomy (ST). After lunch a field training will take place.

On the 3rd and 4th day, there will be a field training. Each day up to 4 profiles will be investigated (up to 2 in the morning and up to 2 in the afternoon).

In each pit, a typical section will be selected and clearly designated as the control section by the contest officials. A measuring tape will be placed in the control section and will be maintained by official pit monitors. The control section must be used for measurement of layer depths and boundaries and for all percentage estimations; it will constitute the officially scored profile and must remain undisturbed and unblocked. Taking samples for studying soil characteristics is only allowed outside the control section. A **pitcard** at each site will give the profile depth to be considered, the number of layers to be described, and chemical or physical data that may be required for classification.

Every participant will receive 1 **scoresheet** per pit (for the group contest only 1 scoresheet per group). Following the instructions of organizers and coaches, the scoresheets are to be filled considering the laboratory information and the instructions of this handbook. There are different national traditions in how to write figures, especially how to differentiate 1 and 7. We respect them all, but in one scoresheet the use must be unambiguous. Teams will be randomly assigned a team number at registration. Individual competitors will be assigned a number that will be used to identify their scoresheet and the rotation schedule. For every training pit, the team coaches will receive the full laboratory dataset and the already filled scoresheets that can be used for guidance.

Conduct of the Individual Contest

Seventy-five minutes will be allowed for evaluating each soil and site for individual judging.

The competitors will be randomly divided into two cohorts. At the first site, cohort 1 will follow this schedule:

1. 5 minutes in the pit,
2. 5 minutes out,
3. 10 minutes in,
4. 10 minutes out,
5. 10 minutes in,
6. 10 minutes out, and
7. 25 minutes free-for-all to enter the pit.

Cohort 2 will follow the opposite in-and-out schedule. At the second site, the cohorts will switch the in-and-out schedule, with cohort 2 in the pit first. Competitors may obtain a sample from the surface layer while out of the pit, as long as they do not enter the pit or disturb those already in the pit. These procedures may be altered prior to the contest to meet unanticipated difficulties at the site.

General rules of the individual contest:

- Competitors must use official codes found in this handbook;
- Competitors are not allowed to speak to each other;

- Competitors are not allowed to use mobile phones, tablets, PDAs or any communication device but may have them on hand in case of health emergency;
- Competitors may have health-related items with them during the contest (such as inhalers or allergy medicine);
- Competitors are allowed to use the equipment provided on site, and the allowed reference materials.
- Each competitor will describe two soil pits. The final score of each competitor will be the total of the two individual-judged pits. No scores are dropped.

Conduct of the Group Contest

Seventy-five minutes will be allowed for evaluating each soil and site for group judging. The time will be divided into segments similar to the individual contest. Groups that start in at the first pit will start out at the next pit. All competitors of each group may participate in the group contest.

General rules of the group contest are the same as of the individual contest, with these exceptions:

- Group members are allowed to speak to other group members (only within the group), as long as their conversation is not loud enough to be heard by other groups;
- Each group will describe two soil pits. The final result for the group contest will be the total of the two group-judged pits. No scores are dropped.

Overall Team Score

For the overall team score only the best three scores at each individually judged pit will be considered for the team final score (this is done so a team of three students can compete). If a team has four members then the lowest score at each pit is dropped. Both group-judged pit scores will count. The final overall team score will consist of the top six individual scores plus the two group scores.

Code of Conduct for the Contest

The host and the IUSS are committed to providing a safe, educational, and productive environment for everyone present at the Soil Judging Contest and therefore prohibit intimidating, threatening, or harassing conduct during the soil judging event. This policy applies to everyone associated with the contest. Participants violating these rules may be asked to leave the meeting at the discretion of contest administrators. Harassment of any kind is a serious issue, will not be tolerated in any form, and includes the following: Offensive comments related to race, ethnicity, religion, disability, age, economic status, physical appearance, gender, or sexual orientation. Deliberate intimidation, stalking, following, unsolicited and harassing photography or recording, sustained disruption of talks or other events, inappropriate physical contact, and unwelcome attention, or any other activity deemed to be harassing will not be tolerated. Participants asked to stop any harassing behavior are expected to comply immediately. If you are being harassed, notice someone else being harassed, or have any other concerns, please contact the contest host who will work with the appropriate leadership to resolve the situation.

1 SITE DESCRIPTION

Task on the scoresheet: Record the profile ID, contestant ID, the number of layers to describe, the bottom depth of the description, and the depth of the nail(s) and the layer number(s) at the bottom of which the nail is/nails are placed. This information will be provided on the pitcard. Additional data will include: lab data as needed, information about cracks or linear extensibility, and presence of heavy metals, pollutants, or pesticides in the soil.

1.1 Land Use

Task on the scoresheet: Indicate the land use type according to Table 1.

Table 1. Land use type (IUSS Working Group WRB, 2022, Table 8.9, modified)

Land-use type	Code
Perennial crop production (e.g. food, fodder, fuel, fiber, ornamental plants)	CPP
Annual crop production (e.g. food, fodder, fuel, fiber, ornamental plants)	CPA
Fallow, less than 12 months, with spontaneous vegetation	FYO
Fallow, at least 12 months, with spontaneous vegetation	FOL
Fallow, all plants constantly removed (dry farming)	FDF
Pasture on unmanaged vegetation	GNP
Intensively managed grassland, pastured	GIP
Intensively managed grassland, not pastured	GIN
Forestry	FOR
Orchard	ORC
Simultaneous agroforestry system	AFS
Nature protection	NPR
Settlement, industry	SEI
Military area	MAR
Other land uses (specified by the contest host)	OLU
Not used and not managed	Ø

1.2 Landform

Task on the scoresheet: Determine the landform type based on Table 2. Some landform choices are shown in Figure 1, Figure 2, and Figure 3. A topographic map with the pit location marked may also be provided.

Table 2. Landform types

Landform	Code
Floodplain (alluvium; low lying area that floods, minimal soil development)	FP
Stream Terrace (alluvium in some part; rarely or non-flooded areas, former floodplains)	ST
Alluvial Fans (alluvium; cone-shaped deposit at tail of steep drainageway)	AF
Drainageway (any parent material; sloping concave area carrying runoff briefly after rains to lower floodplains)	DQ
Anthropogenic (human-transported material deposit or an excavated area)	AN
Upland (none of the other choices)	UP

1.3 Slope Position

Task on the scoresheet: Determine the position according to Figure 1, Figure 2, and Figure 3. Use the codes indicated in Table 3. Slope stakes will be placed on the same landscape position as the pit whenever possible. The area of the slope stakes should be used to determine slope position.

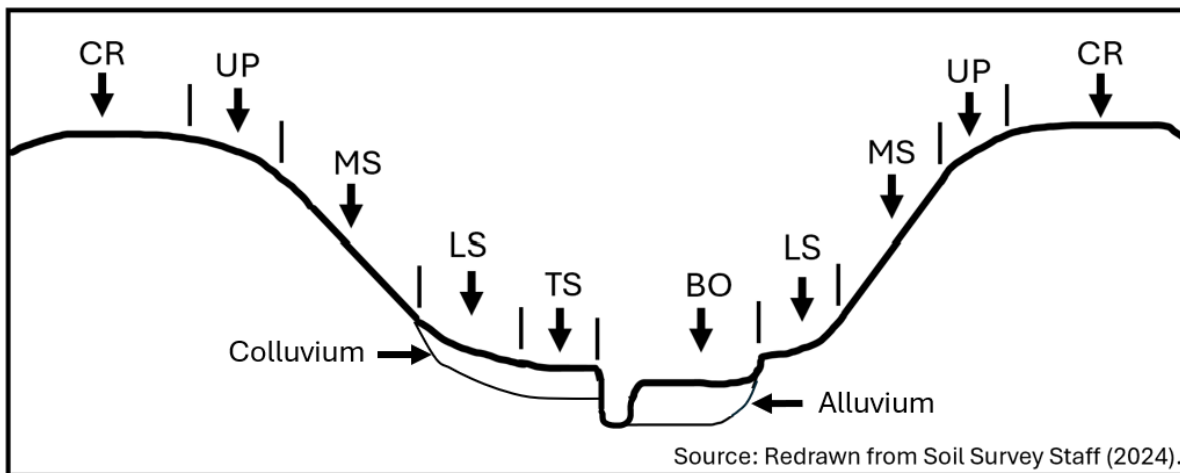


Figure 1. Slope position in undulating and mountainous terrain (Soil Survey Staff, 2024, 1-7, modified)

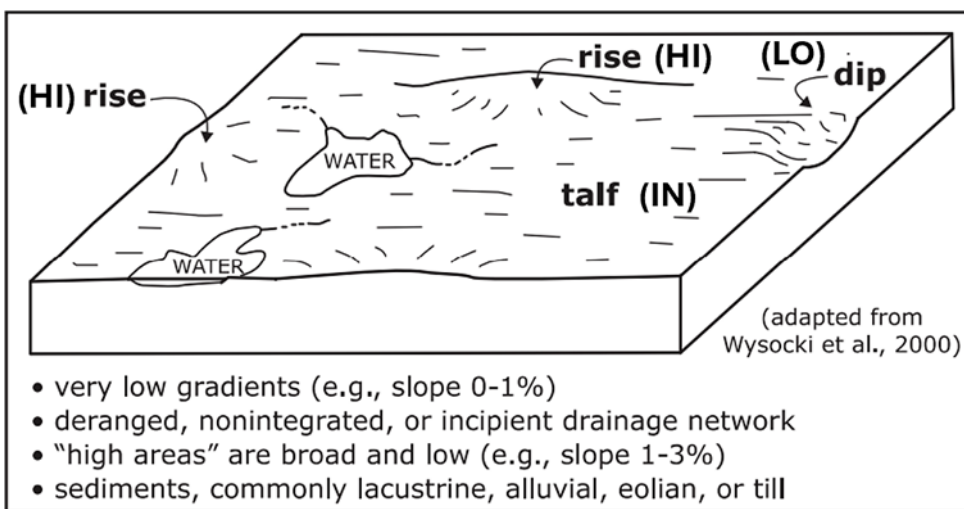


Figure 2. Slope position in flat/almost flat terrain (Soil Survey Staff, 2024, 1-9)

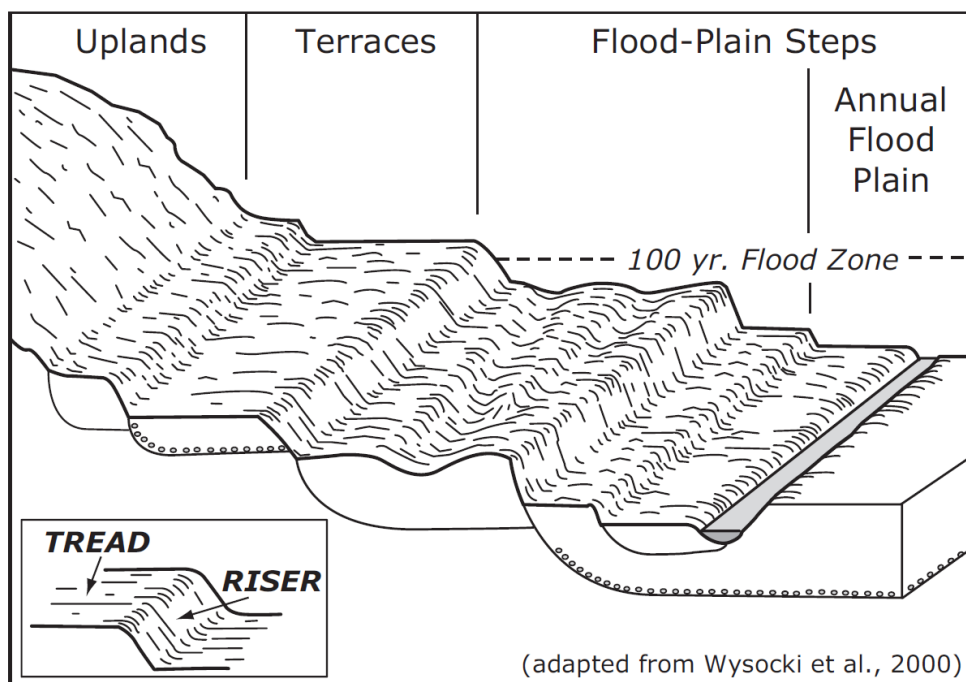


Figure 3. Slope position on natural or human-made terraces/stepped terrain (Soil Survey Staff, 2024, 1-8)

Table 3. Slope position in various terrains (IUSS Working Group WRB, 2022, Table 8.1; Soil Survey Staff, 2024, 1-6 to 1-9)

Position in undulating/mountainous terrain		Position in flat/almost flat terrain	
Landform	Code	Landform	Code
Crest (summit)	CR	Higher part (rise)	HI
Upper slope (shoulder)	UP	Intermediate (talus)	IN
Middle slope (backslope)	MS	Lower part (dip)	LO
Lower slope (foot slope)	LS	Bottom (floodplain)	BO
Toe slope	TS		
Bottom (floodplain)	BO		
Position on terraces or stepped terrain			
Landform		Code	
Riser (sloping ground between treads)		RE	
Tread (flat terrace surface)		TR	

1.4 Slope Gradient

Task on the scoresheet: Determine the slope gradient as a percentage. Slope stakes will be placed on the same landscape position as the pit.

1.5 Rock Outcrops and Surface Rock Fragments

Task on the scoresheet: Estimate on the soil surface the percent of the surface area of rock outcrops and larger rock fragments (also called larger coarse fragments, see *Section 2.3*), both with a diameter > 2 cm. Competitors can refer to the percentage charts of *Section 10*. An area for evaluation will be marked.

1.6 Human-Made Surface Unevenness

Task on the scoresheet: Report the type of human-made surface unevenness with an average height difference of ≥ 5 cm (*Table 4*). Report only if it shows a repeating pattern of two or more. An area for evaluation will be designated by the contest host or indicated on the pitcard.

Table 4. Human-made surface unevenness (IUSS Working Group WRB, 2022, Table 8.20, modified)

Type	Code
Human-made terraces	HT
Raised beds	RB
Drainage canals	CD
Irrigation canals	CI
None	Ø

1.7 Litter Layer

Task on the scoresheet: Report the percentage of the area covered by a litter layer and the average thickness of the litter layer in cm. Competitors can refer to the percentage charts of *Section 10*. An area for evaluation will be marked.

A litter layer (WRB) or a layer of freshly fallen plant litter (ST) is a loose layer that contains > 90% (by volume, related to the fine earth plus all dead plant residues) recognizable dead plant tissues (e.g. undecomposed leaves). The **soil surface (0 cm)** is by convention the surface of the soil after removing, if present, the litter layer. The **mineral soil surface** is the upper limit of the uppermost layer consisting of mineral material (WRB) or mineral soil material (ST) (see *Section 2*, General Definitions).

2 SOIL DESCRIPTION

General Definitions

1. The **fine earth** comprises the soil constituents ≤ 2 mm. The natural mineral constituents > 2 mm are called **rock fragments (coarse fragments)**. The **whole soil** comprises fine earth, rock fragments, artefacts, cemented parts and dead plant residues of any size.
2. In both systems (WRB and ST), a **soil layer** is a horizontal zone in the soil, approximately parallel to the soil surface, with properties different from layers above and/or below it. In ST, the terms layer and horizon are used interchangeably. In WRB, the term soil layer comprises all horizontal zones, whether or not they underwent soil-forming processes. For simplification, in the following, we will use the word layer for horizontal zones and the word horizon when referring to diagnostic zones comprising one or more layers (e.g., argic horizon).
3. **Organic material** in WRB has $\geq 20\%$ (by mass) organic carbon. Organic soil material in ST has $\geq 12\%$ (by mass) organic carbon. The carbon contents or **mineral material** (WRB) and mineral soil material (ST) are below these limit values.
4. For some characteristics, concentrations (accumulations) must be reported. The definitions are given in *Table 5*.
5. The term *dominant* means the most abundant (making up the highest percentage).

Table 5. Types of concentrations (IUSS Working Group WRB, 2022, Table 8.51)

Description	Designation
Rounded body, at least very weakly cemented, that can be removed as discrete unit, with internal organization in the form of concentric layers visible to the naked eye	Concretion
Rounded body, at least very weakly cemented, that can be removed as discrete unit, without evident internal organization	Nodule
Longitudinal body of any cementation class	Filament
Non-cemented or extremely weakly cemented body of variable shape that cannot be removed as discrete unit	Mass
Covering the surfaces of coarse fragments, remnants of a broken-up cemented layers, aggregates or pore walls	Coating

Description Procedure

The minimum thickness of any layer to be described in this context is 5 cm for an A or E layer, 2.5 cm for a Bh, Bsh, Bhs, or Bs layer, 1 cm for a pedogenetically-cemented layer or for a layer with $\geq 12\%$ organic carbon, and 10 cm for all others. If a bedrock layer is present, the organizers try to expose it for at least 10 cm. If there are thinner layers to be described, it will be indicated on the pitcard.

The following characteristics in *Section 2* must be completed for each mineral layer. Only the layer boundaries, rock fragments and artefacts, and carbonates must be described for both organic and mineral layers. Sampling of organic layers should be kept to a minimum (egg-sized), especially for very thin layers. For the decision, which tasks must be completed, the ST definition of organic/mineral layers is used.

2.1 Layer Boundaries

Task on the scoresheet: Determine the *depth (cm)* from the soil surface to the lower boundary of each layer. Determine the *distinctness* of the layer boundaries according to *Table 6* and the *shape (topography)* of the layer boundaries according to *Table 6* and *Figure 4*. *Distinctness* and *shape (topography)* are not described for the lowest layer.

Table 6. Layer boundaries by distinctness and shape (IUSS Working Group WRB, 2022, tables 8.30 and 8.31, modified)

Distinctness			Shape (topography)		
Transition within (cm)	Class	Code	Determination	Class	Code
≤ 2	Abrupt	A	Nearly plane surface	Smooth	S
> 2 to 5	Clear	C	Pockets less deep than wide	Wavy	W
> 5 to 15	Gradual	G	Pockets more deep than wide	Irregular	I
> 15	Diffuse	D	Discontinuous	Broken	B

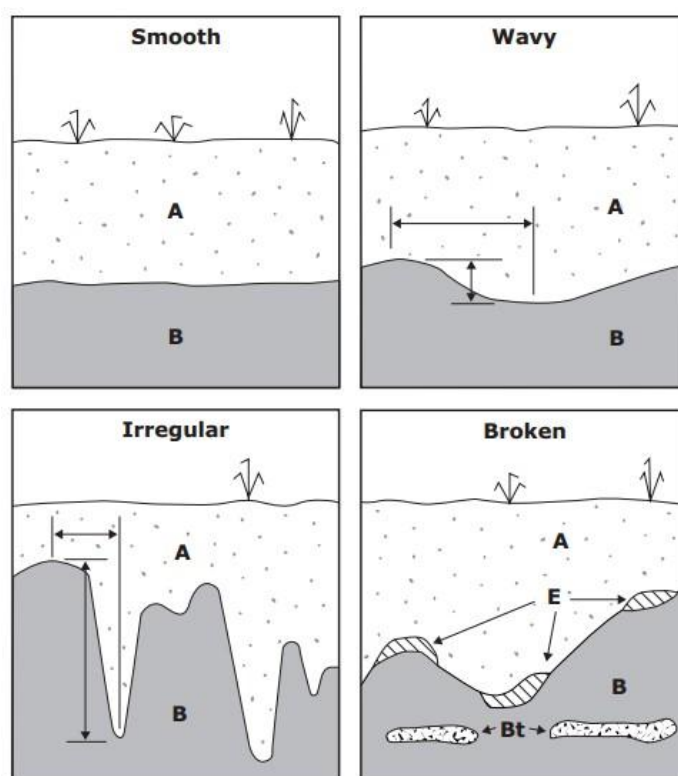


Figure 4. Shape (topography) of layer boundaries (Soil Survey Staff, 2024, 2-7)

2.2 Texture

Task on the scoresheet: Determine the texture class of the fine earth (≤ 2 mm) for each mineral layer using the texture triangle and the codes indicated there (*Figure 5*). You may use the provided flow chart (*Figure 6*) or any other scheme for determining texture. In addition to the texture class, a field estimate of the percentage of clay and sand must be given. This estimate is useful for indicating increases or decreases in clay and/or sand content within texture classes, and for comparing field estimates with analytical results.

In different countries, different limit values for particle-size classes and different texture triangles are used. This contest uses the definitions of WRB and ST (*Table 7*). The limit values for differentiating sand and silt are slightly different between the two systems, but this should not have a significant effect on hand texturing.

Table 7. Particle-size classes according to Soil Taxonomy (Soil Survey Staff, 2022) and WRB (IUSS Working Group WRB, 2022)

Particle-size class	Soil Taxonomy	WRB
Sand	> 50 μm - $\leq$ 2 mm	> 63 μm - $\leq$ 2 mm
Silt	> 2 μm - $\leq$ 50 μm	> 2 μm - $\leq$ 63 μm
Clay	$\leq$ 2 μm	$\leq$ 2 μm

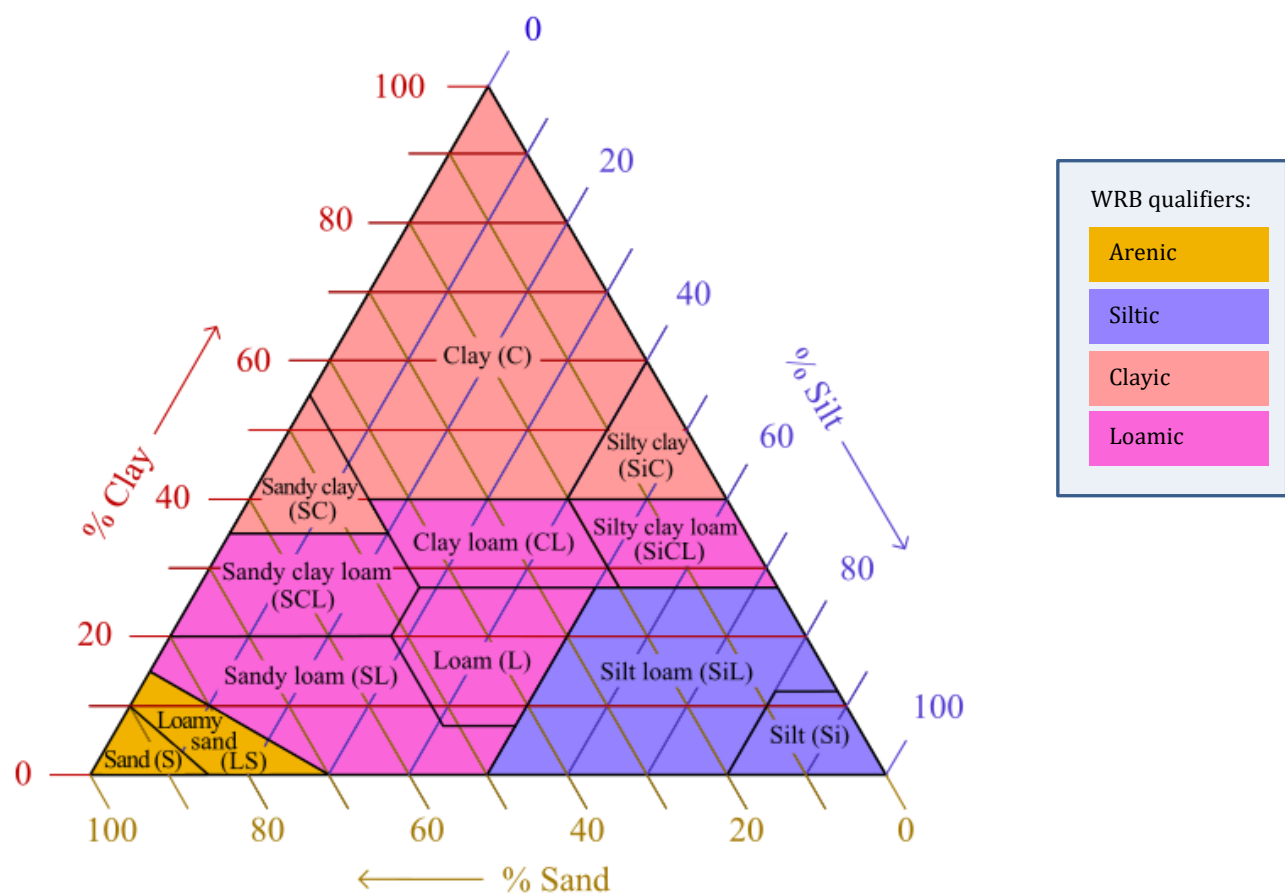


Figure 5. Texture classes with codes and with the correlation with the WRB texture qualifiers (IUSS Working Group WRB, Figure 8.13)

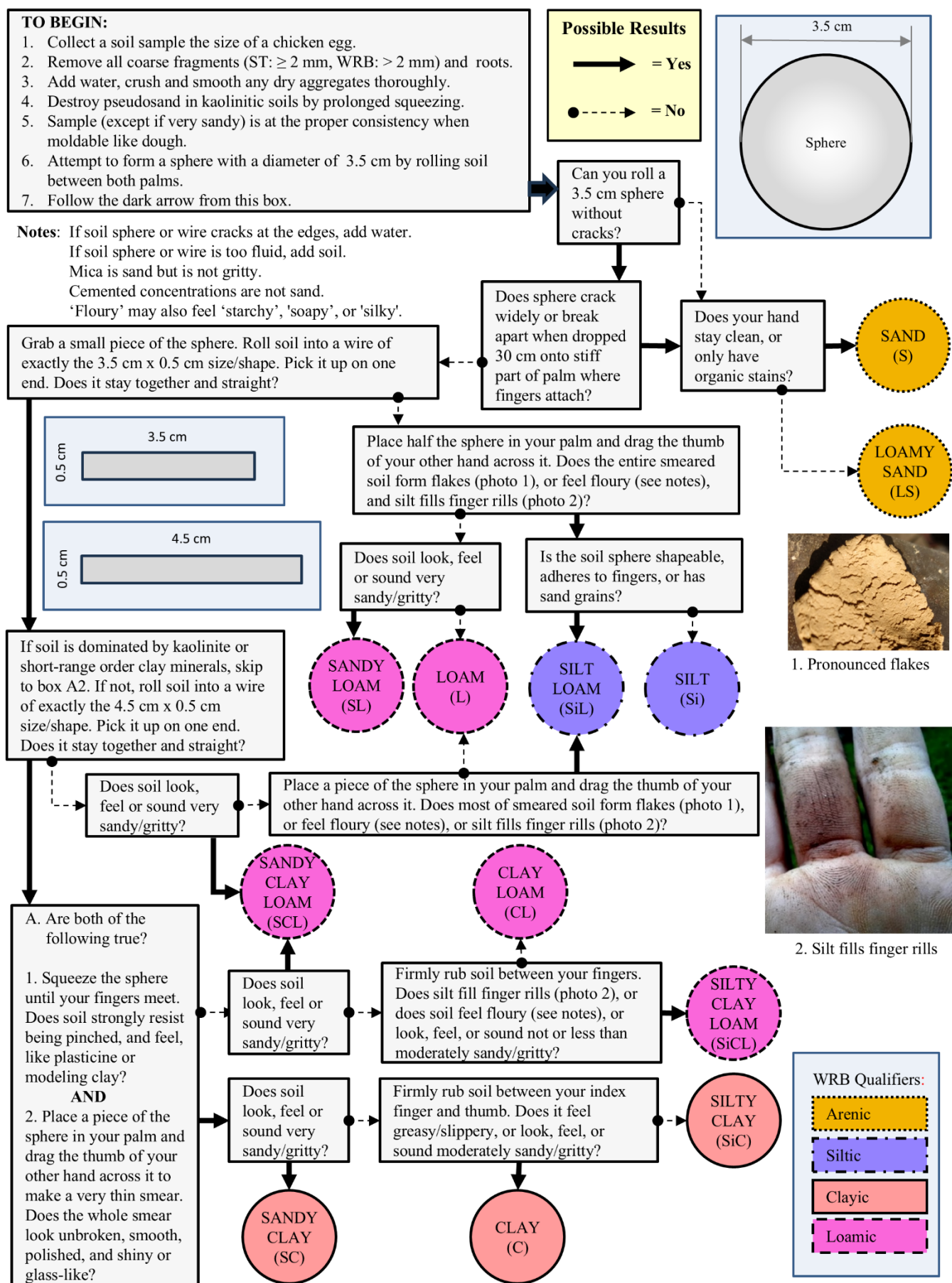


Figure 6. Flow chart for determination of texture classes (IUSS Working Group WRB, Figure 8.14, modified)

2.3 Rock Fragments and Artefacts (R.F./AF)

Task on the scoresheet: Estimate the abundance of rock fragments (also called coarse fragments) by volume in % for each layer. Rock fragments are unattached pieces of geologic origin, > 2 mm, that do not fail under foot pressure by full body weight. Competitors can refer to the percentage charts of *Section 10* for estimating percentages. If artefacts > 2 mm are present, they are included in this estimation. After estimating the abundance for every layer, a weighted average must be calculated from the mineral soil surface to 100 cm or to continuous rock or to a cemented layer with cementation class of at least moderately cemented, whichever is shallower. Information about cementation will be given on the pitcard, where necessary.

2.4 Structure

Task on the scoresheet: Record the dominant structure type for each mineral layer using *Table 8* and *Figure 7*. Record the dominant grade of structure for each mineral layer using the codes indicated in *Table 9*. If the structure type is SGR, MA, STR or CDY, the structure grade must be $\emptyset$. The grade of structure can be judged both on how evident the structural units are in place before extraction and how well they hold together after extraction and being gently tumbled.

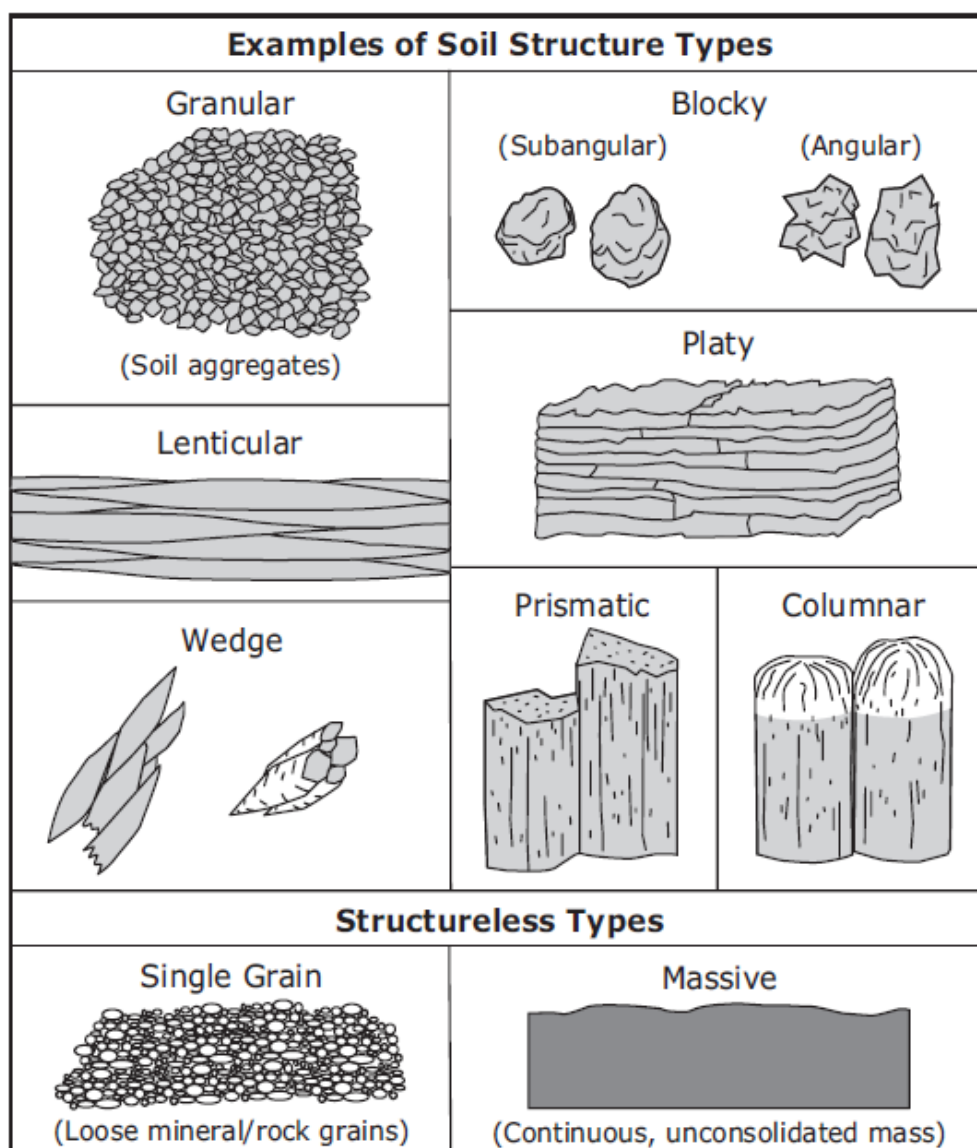


Figure 7. Examples of soil structure types (Soil Survey Staff, 2024, 2-54)

Table 8. Soil structure types (Soil Survey Staff, 2024, 2-53, modified)

Type	Determination	Code
NATURAL SOIL STRUCTURAL UNITS (pedogenic structure)		
Granular	Small polyhedrals with curved or very irregular faces.	GR
Angular blocky	Polyhedrals with faces that intersect at sharp angles (planes).	ABK
Subangular blocky	Polyhedrals with subrounded and planar faces lacking sharp angles.	SBK
Platy	Flat and platelike units.	PL
Wedge	Elliptical, interlocking lenses that terminate in acute angles.	WEG
Prismatic	Vertically elongated units; flat tops.	PR
Columnar	Vertically elongated units with rounded tops that commonly are "bleached."	COL
STRUCTURELESS		
Single grain	No structural units; entirely noncoherent; e.g., loose sand.	SGR
Massive	No structural units; material is a coherent mass (not necessarily cemented).	MA
Stratified	No structural units; visible stratification from sedimentation	STR
ARTIFICIAL SOIL FRAGMENTS OR CLODS (nonpedogenic structure)		
Cloddy	Irregular blocks created by artificial disturbance; e.g., tillage or compaction.	CDY

Table 9. Grades of aggregates (IUSS Working Group WRB, Table 8.43, modified)

Determination	Grade	Code
Aggregates are barely observable in place, and there is only a weak arrangement of natural surfaces of weakness. When gently disturbed, the soil material breaks into a mixture of few entire aggregates, many broken aggregates, and much loose material that is not in aggregates. Aggregate surfaces differ in some way from aggregate interiors.	Weak	WE
Aggregates are observable in place, and there is a distinct arrangement of natural surfaces of weakness. When disturbed, the soil material breaks into a mixture of many entire aggregates, some broken aggregates, and little loose material that is not in aggregates. Aggregate surfaces generally show distinct differences with aggregate interiors.	Moderate	MO
Aggregates are clearly observable in place, and there is a prominent arrangement of natural surfaces of weakness. When disturbed, the soil material breaks mainly into entire aggregates. Aggregate surfaces generally differ markedly from aggregate interiors.	Strong	ST

2.5 Soil Color

Task on the scoresheet: Use the Munsell® color chart to determine the dominant color of each mineral layer (i.e., the color that occupies the largest volume (ST) or the largest exposed area (WRB)). In some layers the color of a reductimorphic or an oximorphic feature, given on the pitcard (see *Section 2.7*) may be the dominant color. The color of clay coatings (see *Section 2.8*) must not be considered for determining the dominant color. Colors must be designated by Hue, Value and Chroma. For this contest, Chroma unit 5 and 7, not indicated on the charts, must be used where they are a better choice than chroma chips 4, 6 or 8. Besides that, only the colors indicated on the charts are allowed.

Take a fresh sample, slightly crush it, if loose, and observe the color. The moist state corresponds to field capacity, which is obtained with sufficient accuracy by moistening and reading the color as soon as visible moisture films have disappeared.

2.6 Carbonates

Task on the scoresheet: Estimate the CaCO_3 content of the fine earth of each layer by adding some drops of 10% HCl to the soil. Using *Table 10*, match the reaction to the criterion that best fits and record the appropriate code. Indicate the dominant type of secondary carbonates according to *Table 11* with the help of *Table 5*. Additionally, report the total exposed area covered by secondary carbonates of any type. Report it in % related to the fine earth plus accumulations of secondary carbonates of any size and any cementation class. For this contest, the exposed area is considered to be equivalent to the volume.

Table 10. Carbonate contents in the soil matrix (IUSS Working Group WRB, 2022, Table 8.62)

Criterion	Content	% (by mass)	Code
No visible or audible effervescence	Non-calcareous	0	Ø
Audible effervescence but not visible	Slightly calcareous	> 0 to 2	SL
Visible effervescence	Moderately calcareous	> 2 to 10	MO
Strong visible effervescence, bubbles form a low foam	Strongly calcareous	> 10 to 25	ST
Extremely strong reaction, thick foam forms quickly	Extremely calcareous	> 25	EX

Table 11. Types of secondary carbonates (IUSS Working Group WRB, 2022, Table 8.64)

Type	Code
Masses (including spheroidal aggregations like white eyes (byeloglaska); including masses filling the complete fine earth)	MA
Nodules and/or concretions	NC
Filaments (including continuous filaments like pseudomycelia)	FI
Coatings on soil aggregate surfaces or biopore walls	AS
Coatings on undersides of coarse fragments and of remnants of broken-up cemented layers (with or without coatings on other sides)	UR
No secondary carbonates	NO

2.7 Redoximorphic Features

Task on the scoresheet: In some layers, a reductimorphic or an oximorphic feature makes up the dominant color (see *Section 2.5*). In other layers, redoximorphic features could make up a minor amount or be entirely absent. If present, the colors of the dominant oximorphic feature (redox concentration) and the dominant reductimorphic feature (redox depletion) will be given on the pitcard. If no colors are given on the pitcard, reductimorphic and oximorphic features should not be recorded, even if some are present. If the dominant reductimorphic feature color in a layer is not value ≥ 4 and chroma ≤ 2 but there are some reductimorphic features with those colors present, the color of those non-dominant features will be given on the pitcard in addition to a dominant reductimorphic feature color. Mn oxide coatings are ignored in this contest.

Indicate the locations of the **dominant** oximorphic feature and the **dominant** reductimorphic feature for each mineral layer according to *Table 12* and *Table 13* with the help of *Figure 8*. Report the combined abundance of OOA plus OOE. Then, report the total abundance of **all** oximorphic features and **all** reductimorphic features. Report all abundancies as a percentage of the exposed area of the entire layer as seen on the profile wall (related to the fine earth plus oximorphic features of any size and any cementation class). For estimating the percentages, the profile wall may be sprayed with water. If redox features are absent or not recorded, write Ø in all relevant cells in the scoresheet.

Redoximorphic features (RMF) are soil morphological features caused by reduction processes or by reduction and subsequent oxidation processes. Most RMFs are the result of the reduction/oxidation of iron (Fe) and, to a lesser extent, manganese (Mn), minerals. Iron is a major pigment that influences soil color. The loss, accrual, and valence/mineral state of Fe are major determinants of color patterns within or across soil layers. Microorganisms reduce iron or manganese when free oxygen is limited or excluded from a soil volume or layer by water saturation for extended time. Reduced iron (Fe^{2+}) and manganese (Mn^{2+}) are comparatively much more soluble and mobile than oxidized iron (Fe^{3+}) and manganese (Mn^{4+}) and move with water flow and due to redox gradients. Reduced Fe and Mn are oxidized and precipitate when water drains from soil (re-entry of free oxygen), or in parts where oxygen is still present. In soils with stagnant water, this is inside aggregates and in soils with ascending groundwater along soil pores, including root channels, or along roots. The re-oxidized Fe or Mn form concentrations like *masses*, *concretions* or *nodules* (see Table 5). Inside one aggregate, several masses and/or concretions and/or nodules may occur. Concretions and/or nodules may even occur within masses. Figure 8 shows the location of oximorphic features. The corresponding reductimorphic features look similar but just inverse. In WRB, reductimorphic and oximorphic features are defined in relation to each other.

Zones in the soil with oxidized Fe present often show redder and/or more intense (higher chroma) colors than adjacent soil zones, while soil zones with oxidized Mn present often show a darker color than adjacent soil zones. Reduced Fe and Mn are colorless, so reductimorphic features that have only reduced Fe and Mn or no Fe or Mn present are more pale or gray (yellowish and/or lower chroma) than adjacent soil zones.

Table 12. Location of oximorphic features (IUSS Working Group WRB, 2022, Table 8.55, modified)

Location		Code
Inner parts	Inside soil aggregates: masses	OIM
	Inside soil aggregates: concretions and/or nodules	OIC
	Inside soil aggregates: combinations of masses with concretions and/or nodules	OIB
	Dominant color (large masses), including parts inside soil aggregates	OID
Outer parts	On outer parts of soil aggregates	OOA
	On biopore walls	OOP
	Dominant color, including outer parts of soil aggregates and/or biopore walls	OOD
Throughout	Throughout	ORT

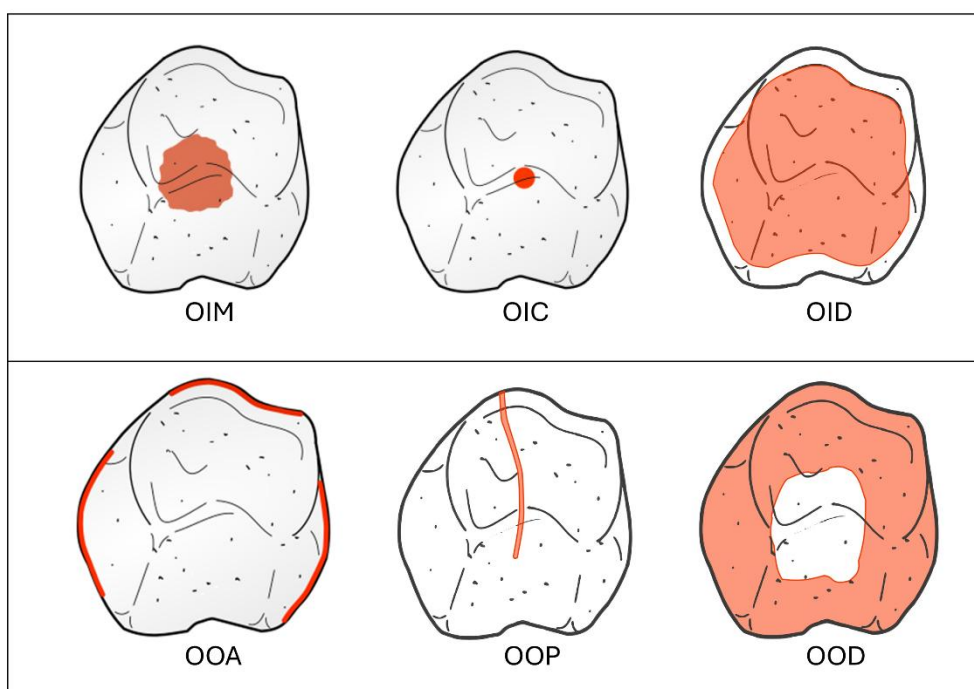


Figure 8. Location of some oximorphic features (IUSS Working Group WRB, 2002, Figure 8.19, modified)

Table 13. Location of redoximorphic features (IUSS Working Group WRB, 2022, Table 8.56, modified)

Location		Code
Inner parts	Zones inside soil aggregates	RIA
	Dominant color, including zones inside soil aggregates	RID
Outer parts	Outer parts of soil aggregates	ROA
	Around biopores	ROP
	Dominant color, including outer parts of soil aggregates and/or zones around biopores	ROD
Throughout	Throughout	RRT

2.8 Aggregate Surface Features

Task on the scoresheet: Determine and record the dominant type of features at the surfaces of aggregates and along biopores, other than redoximorphic features, for each mineral layer, except for plow layers and layers of human-transported material, using *Table 14*.

Clay coatings (*Figure 9*) result from clay migration and show a shiny appearance due to illuviated clay minerals and a more intense color (usually a higher chroma) due to illuviated iron oxides. If sand grains are present, they are waxed over by clay particles. Stress features (*Figure 10*) do not differ in color from the interiors of the aggregates. They form when soil aggregates are pressed against each other due to swelling clay minerals. Striations form when sand grains are moved with strong pressure along the surfaces of aggregates sliding past each other. If striations are present, the stress features are called slickensides, otherwise they are called pressure faces.

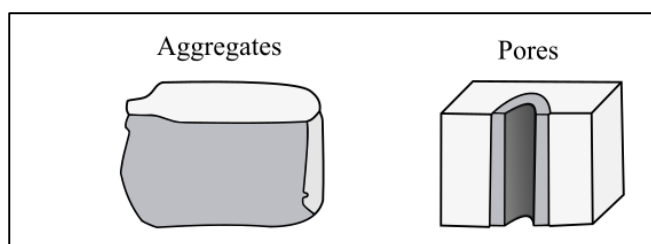


Figure 9. Clay coatings (Soil Survey Staff, 2024, 2-33; IUSS Working Group WRB, 2022, Figure 8.20)

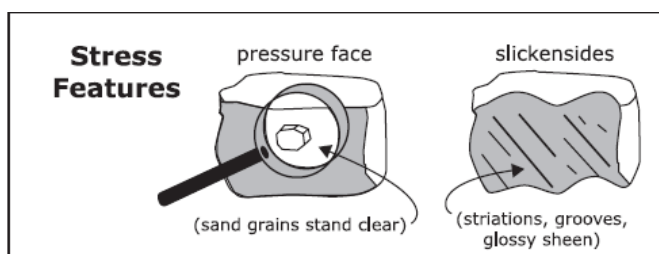


Figure 10. Stress features (Soil Survey Staff, 2024, 2-33)

Table 14. Aggregate surface features

Type and abundance	Code
Clay coatings, < 15% of surfaces of soil aggregates and biopore walls covered	CF
Clay coatings, ≥ 15% of surfaces of soil aggregates and biopore walls covered	CM
Slickensides, < 5% of surfaces of soil aggregates covered	SV
Slickensides, ≥ 5% and < 10% of surfaces of soil aggregates covered	SF
Slickensides, ≥ 10% of surfaces of soil aggregates covered	SM
Pressure faces, any abundance	PF
None	∅

2.9 Parent Materials

Task on the scoresheet: Determine the parent materials for each mineral layer according to Table 15. If the parent material is igneous, metamorphic or consolidated sedimentary rock, indicate only the *major class code*. If the parent material is unconsolidated sedimentary rock, indicate the *group code*.

Table 15. Types of parent materials (IUSS Working Group WRB, 2022, Table 8.91, modified)

Lithology	Major class code	Group code
Igneous rock	I	
Metamorphic rock	M	
Consolidated sedimentary rock	S	
Unconsolidated sedimentary rock: Weathered residuum	U	UR
Unconsolidated sedimentary rock: Fluvial	U	UF
Unconsolidated sedimentary rock: Lacustrine	U	UL
Unconsolidated sedimentary rock: Colluvial	U	UC
Unconsolidated sedimentary rock: Aeolian	U	UE
Unconsolidated sedimentary rock: Glacial	U	UG
Unconsolidated sedimentary rock: Anthropogenic/ Technogenic	U	UA
Unconsolidated sedimentary rock: Human-transported	U	UT

3 LAYER DESIGNATIONS

Teams and individual contestants must give the layer designations according to both Soil Taxonomy (ST, 13th edition, 2022) and World Reference Base for Soil Resources (WRB, 4th edition, 2022). The analytical data required for layer designation will be provided on the pitcard.

In both systems, the layer designations must be completed for both mineral and organic layers. The layer designation consists of four elements:

1. If a soil consists of more than one stratum (WRB: lithic discontinuity, ST: lithologic discontinuity), from the second stratum downwards, every layer designation starts with the **number of the stratum**, indicated with an Arabic numeral (prefix). The first stratum does not receive such a number. In Soil Taxonomy, where appropriate, the caret symbol (^) is added (*Section 3.2*).
2. The next is the **master symbol** (master letter), which is a capital letter indicating the major characteristics of the layer. Transitional and combination layers receive more than one master symbol as explained below.
3. The master symbol may be followed by one or more lowercase letters, called **suffixes** (subordinate letters). For transitional and combination layers, different rules apply in WRB and Soil Taxonomy on how to add the suffix(es) to the master symbols.
4. If two or more layers have an identical combination of master symbol and suffixes, they have to be further differentiated, following different rules in WRB (*Section 3.1*) and Soil Taxonomy (*Section 3.2*).

Transitional layers are dominated by properties of one master layer but having subordinate properties of another. Two master symbols are used, such as AB, EB, BE or BC. The symbol of the material that dominates is given first.

Combination layers have distinct parts and recognizable properties of two kinds of master layers. If they occur side-by-side in the same layer, the two master symbols are separated by a virgule (/), such as A/C, B/E, B/C or C/R. If they occur alternating vertically such as lamellae, they are separated by an ampersand (&). Contestants should only describe the material that represents the first master horizon letter named, for example, the A part of an A/C horizon.

3.1 World Reference Base for Soil Resources

Task on the scoresheet: Indicate the full layer designation. For the master symbol use *Table 16* and for the suffixes *Table 17*. A maximum of three suffixes may be used if applicable.

Specific rules for WRB:

- If organic layers occur and it is not obvious whether they are saturated by water or not, this information is given on the pitcard.
- In transitional and combination layers, every suffix is placed directly after its master symbol (example: BsC, CBs). (Due to the division of Section 3.1 of the scoresheet into columns, this rule is not relevant for the contest.)
- If two or more layers have an identical combination of master symbol and suffixes, they have to be differentiated using Arabic **numerals** (example: A-Bw1-Bw2-2Bw3-C). This applies even if interrupted by other layers (example: A-E1-Bh1-E2-Bh2-C) and across different strata (example: A1- Bw1-2A2-2Bw2-2Bw3).

The sequence for naming layers in WRB is (indicated are the columns of Section 3.1 of the scoresheet):

1. Prefix numerals (first column);
2. Master symbols (second column);
3. Suffixes (third column);
4. Vertical subdivision numerals to differentiate identically named layers (fourth column).

Table 16. Master symbols for layers in WRB (IUSS Working Group WRB, 2022, Table 10.1, modified)

Symbol	Short description
Mineral layers	
A	Mineral layer at the mineral soil surface or buried; contains organic matter that has at least partly been modified in situ; soil structure and/or structural elements created by agricultural practices in $\geq 50\%$ (by volume, related to the fine earth); usually a darker color than underlying layers.
E	Mineral layer, at or near the mineral soil surface, characterized by a loss of one or more of the following: Fe, Al, clay minerals, organic matter; usually lighter in color (higher value and/or lower chroma) than the overlying A and underlying B.
B	Mineral layer that has (at least originally) formed below an A or E layer; rock structure, if present, in $< 50\%$ (by volume, related to the fine earth); one or more of the following processes of soil formation: • formation of soil aggregate structure • formation of clay minerals and/or oxides • accumulation by illuviation processes of one or more of the following: Fe, Al, clay minerals, organic matter, silica, carbonates, gypsum • removal of carbonates or gypsum.
C	Mineral layer, unconsolidated (can be cut with a spade when moist) or consolidated and more fractured than the R layer; may be partly chemically weathered, otherwise little affected by pedogenic processes.
R	Consolidated rock that cannot be cut with a spade; fractures, if present, occupy $< 10\%$ (by volume, related to the whole soil); not resulting from the cementation of a soil layer.
Organic layers	
O	Organic layer ($\geq 20\%$ (by mass) organic carbon), not forming part of a litter layer; O layers are not saturated by water for prolonged periods.
H	Organic layer ($\geq 20\%$ (by mass) organic carbon), not forming part of a litter layer; H layers are saturated by water for prolonged periods or have been drained.

Table 17. Suffixes for layers in WRB (IUSS Working Group WRB, 2022, Table 10.2, modified)

Suffix	Short description	Combination with
b	Buried layer; first, the layer has formed, and then, it was buried by mineral material	H, O, A, E, B
c	Concretions or nodules (only used if following another suffix that indicates the accumulated substance; by carbonates, gypsum, silica, Fe and Al oxides)	
g	Accumulation of Fe oxides predominantly inside soil aggregates or loss of Fe by lateral subsurface flow (as in stagnic properties)	A, B, C E
h	Significant amount of organic matter; in A layers at least partly modified in situ; in B layers predominantly by illuviation; in C layers forming part of the parent material	A, B, C

i	Slickensides and/or wedge-shaped aggregates	B
k	Accumulation of secondary carbonates	H, O, A, E, B, C
l	Accumulation of oxidized Fe (predominantly at soil aggregate surfaces and biopore walls) following transportation in reduced form in upward-moving capillary water (as in gleyic properties)	H, A, B, C
m	Cementation by pedogenetic processes (only used if following another suffix that indicates the accumulated substance; by carbonates, gypsum, silica, soluble salts, Fe and Al oxides)	
p	Modification by cultivation (e.g. ploughing); mineral layers are designed A, even if they belonged to another layer before cultivation	H, O, A
r	Strong reduction (as in gleyic properties)	A, E, B, C
s	Accumulation of Fe oxides and Al by vertical illuviation processes from above	B, C
t	Illuvial accumulation of clay minerals (with or without associated oxides)	B, C
u	Containing artefacts or consisting of artefacts (related to the whole soil)	H, O, A, E, B, C, R
w	Formation of soil aggregate structure and/or oxides and/or clay minerals	B
α	Primary carbonates (in R layers related to the rock, in all others related to the fine earth) and no prominent accumulation of secondary carbonates	H, A, E, B, C, R
δ	High bulk density so that roots cannot enter, except along cracks (natural or anthropogenic - not due to cementation (symbol ..m), not in fragic horizons (symbol x), not in layers with retic properties (symbol Bt/E))	A, E, B, C
τ	Human-transported natural material (related to the whole soil)	H, O, A, E, B, C

3.2 Soil Taxonomy

Task on the scoresheet: Indicate the full layer designation. For the master symbol use *Table 18* and for the suffixes *Table 19*. A maximum of three suffixes may be used if applicable.

Specific rules for Soil Taxonomy:

- The caret symbol (^) is used as a prefix to identify human-transported material. (Designation of a lithologic discontinuity with the material beneath is an independent decision, depending on the source of the transported material).
- In transitional layers, first all master symbols are placed and then all suffixes (example: BCs, BCs). In combinations layers, the suffixes follow the master symbol directly (examples: Bt/E, E & Bt).
- If two or more non-adjacent layers have an identical combination of prefix, master symbol and suffixes, they have to be differentiated using prime symbol that follows the master letter(s) (example: A-E-Bt-E'-B't1-B't2-2Bt-2C).
- If two or more adjacent layers have an identical combination of prefix, master symbol and suffixes, they have to be differentiated using Arabic numerals (example: A-Bw1-Bw2-2Bw-C).

The sequence for naming layers in Soil Taxonomy is (indicated are the columns of Section 3.2 of the scoresheet):

1. Prefix numerals and then the caret symbol (first column);
2. Master letters (second column);
3. Prime to differentiate non-adjacent identically named layers and then the suffixes (subordinate letters) (third column);
4. Vertical subdivision numerals to differentiate adjacent identically named layers (fourth column).

Table 18. Master symbols for layers in ST (Soil Survey Staff, 2024, 4-1 to 4-3, modified)

Symbol	Short description
Mineral layers	
A	Mineral soil material, formed at the surface or below an O layer, little remnant rock structure; one or more of the following: • accumulation of organic matter • properties resulting from cultivation • morphology resulting from surficial processes; usually a darker color than underlying layers.
E	Mineral layer, at or near the soil surface, characterized by a loss of one or more of the following: Fe, Al, clay minerals, organic matter; usually lighter in color (higher value and/or lower chroma) than the overlying A and underlying B.
B	Mineral layer that has (at least originally) formed below an A or E layer; little or no rock structure; one or more of the following processes of soil formation: • formation of soil aggregate structure • formation of clay minerals and/or oxides • accumulation by illuviation processes of one or more of the following: Fe, Al, clay minerals, organic matter, silica, carbonates, gypsum • removal of carbonates or gypsum.
C	Mineral soil material; soft or weakly to moderately coherent or highly cracked bedrock (can be cut with a spade when moist); may be partly chemically weathered, otherwise little affected by pedogenic processes.
R	Hard bedrock (continuous, coherent, at least strongly coherent) that cannot be cut with a spade; not resulting from the cementation of a soil layer.
Organic layers	
O	Organic layer ($\geq 12\%$ (by mass) organic carbon), not forming part of freshly fallen plant litter.

Table 19. Suffixes for layers in ST (Soil Survey Staff, 2024, 4-3 to 4-6, modified)

Suffix	Short description	Used for
b	Buried pedogenetic layer	O, A, E, B
c	Concretions or nodules	A, E, B
d	Physical root-limitation, high bulk density (natural or human-induced)	C
g	Strong gleying (depleted matrix)	A, E, B, C
h	Illuvial and synthesized Al-humus complexes (as in spodic materials)	B
k	Accumulation of secondary carbonates	A, B
kk	Accumulation of $\geq 50\%$ pedogenic carbonates, soil fabric is plugged	B
m	Cementation by pedogenic processes	B
n	Pedogenic accumulation of exchangeable sodium	B
p	Plowing or other human mixing disturbance	O, A
r	Paralithic bedrock	C
s	Illuvial and synthesized oxides of Fe, Mn, or Al (as in spodic materials)	B
t	Illuvial accumulation of clay	B
u	Presence of human-manufactured materials (artifacts)	H, O, A, E, B C
w	Development of color or structure, absence of more advanced genesis	B
x	Fragic character (brittle nature of rupture)	B
y	Accumulation of secondary gypsum	A, B
z	Pedogenic accumulation of salts more soluble than gypsum	A, B, C

3.3 Major differences in naming layers in WRB and Soil Taxonomy

Table 20. Major differences in naming layers in WRB and Soil Taxonomy.

Item	WRB	Soil Taxonomy
Vertical subdivision numbering	Adjacency does not matter	Adjacency matters
Water-saturated organic layers	H	O
r	Reductimorphic features are the dominant color	Paralithic bedrock
g	Stagnic properties are present	The dominant color is value ≥ 4 and chroma ≤ 2 from reduction processes
h	Significant amount of organic matter	Accumulation of illuvial organic matter (spodic materials)
Slickensides	i	ss
Dense material	δ (delta)	d
Human-transported material	τ (tau)	^ (caret)

4 SOIL PROFILE CHARACTERISTICS

For each of the seven characteristics to be assessed, competitors are to place an 'X' in one box only. Depth values are related to the mineral soil surface (see *Section 1.7*), unless stated otherwise

4.1 Type of Restrictive Layer

Task on the scoresheet: Determine the type of restrictive layer.

Root- and/or water-restrictive layers include:

- Bedrock (ST: lithic (R) or paralithic (Cr) materials, WRB: continuous rock (R));
- Cemented layers with the layer suffix m (both ST and WRB);
- Densic materials with physical root-restriction and high bulk density (natural or human-induced) with the layer suffix d (ST) or δ (WRB);
- Layers with texture classes of SiC, C or SC that are structureless and massive.

4.2 Effective Soil Depth

Task on the scoresheet: Determine the type of the effective soil depth class.

Effective soil depth is defined as the depth from the mineral soil surface to the upper boundary of a restrictive layer. Use the classes in *Table 21*. If the lower depth of judging is ≤ 120 cm, and there is no restrictive layer within or at the judging depth, the layer encountered at the bottom of the judged profile may be assumed to continue to > 120 cm and 'very deep' should be selected.

Table 21. Effective soil depth

Depth	Depth class	Code
≤ 20 cm	Very shallow	VS
> 20 to 40 cm	Shallow	S
> 40 to 80 cm	Moderately deep	MD
> 80 to 120 cm	Deep	D
> 120 cm	Very deep	VD

4.3 Hydraulic Conductivity

Task on the scoresheet: Determine the hydraulic conductivity class of the *mineral surface layer* and of the *layer with the lowest conductivity*.

Critical for agronomic soil functions and partitioning of rainfall, we will estimate the saturated hydraulic conductivity of the *mineral surface layer* and of the *layer with the lowest conductivity* within the depth specified on the pitcard.

Three general hydraulic conductivity classes are used:

High (H) – high hydraulic conductivity class is indicated with any of the following:

- Layers with texture classes of *sand* or *loamy sand*.
- *Sandy loam*, *sandy clay loam*, *silt loam* and *loam* texture classes that are especially 'loose' because of very high organic matter content (> 5% organic carbon).
- Layers containing > 60% of coarse fragments with insufficient fine earth to fill voids between fragments.

Moderate (M) – this includes those materials excluded from 'low' and 'high' classes.

Low (L) – low hydraulic conductivity class is indicated with any of the following:

- Layers with texture classes of *clay*, *silty clay* or *sandy clay* and one or more of the following: (1) slickensides or pressure faces, (2) structure grade of WE, (3) platy or massive structure type, or (4) structure grade of MO and reductimorphic features.
- Layers with texture classes of *silty clay loam* or *clay loam* that have reductimorphic features and a structure grade of WE or are massive.
- Layers with one of the following dominant structures: (1) columnar structure, (2) prismatic structure with a diameter of ≥ 10 cm, (3) blocky structure with a diameter of ≥ 5 cm. These structural units do not part into sub-structural units with a grade of MO or WE.
- Dense layers with the layer suffixes d (ST) or δ (WRB).
- Cemented layers with the layer suffix m (both ST and WRB).
- Bedrock layers (ST: lithic or paralithic materials, WRB: continuous rock) where the layer directly above contains reductimorphic features.

4.4 Available Water-Holding Capacity (AWC)

Task on the scoresheet: Calculate the available water for each non-restrictive layer using the formula: Thickness x Available Water Capacity x [(100-% coarse fragments)/100].

Sum the available water of all non-restrictive layers and portions of non-restrictive layers to 150 cm. This sum is the available water-holding capacity of the soil.

Critical to agronomic interpretations for crop growth, the available water-holding capacity (also called:

available water capacity) is approximately the water held between field capacity (pF 1.8) and permanent wilting point (pF 4.2). The approximate amount of available water stored in the soil is calculated for the top 150 cm of the soil profile, but not below the top of a restrictive layer (*Section 4.1*). If the profile is described to less than 150 cm and there are no restrictive layers, the lowest described layer is assumed to extend to 150 cm. If the lowest described non-restrictive layer extends below 150 cm, its lower depth is reduced back to 150 cm for available water calculation purposes. Texture is an important factor influencing available water capacity. The relationship between texture classes and available water retained per centimeter of soil is given in *Table 22*.

Table 22. Relationship between texture classes and available water capacity

Texture classes	Available Water Capacity (cm water per cm soil)
sand, loamy sand	0.07
sandy loam	0.10
sandy clay loam, sandy clay, loam, clay loam, clay, silty clay	0.15
silt, silt loam, silty clay loam	0.20
layers with $\geq 12\%$ organic carbon	0.50

(Note: The actual AWC values for sand, loamy sand and sandy loam classes depend on the size of the sand grains; coarse sand has a smaller AWC than fine sand, but this is not accounted for in this contest. Also, the AWC of organic layers is highly variable and decreases with advancing decomposition; the indicated value of 0.5 cm/cm is a rough mean value.)

Coarse fragments are considered to have negligible (assume zero) water retention. If a soil contains coarse fragments, the volume occupied by the coarse fragments must be estimated and the available water-holding capacity corrected accordingly.

For example, if a silt loam A layer is 25 cm thick and contains coarse fragments which occupy 10% of its volume, the available water of this layer would be

$$25 \text{ cm} \times 0.20 \text{ cm/cm} \times [(100-10)/100] = 4.50 \text{ cm.}$$

Four classes of available water-holding capacity will be used (*Table 23*):

Table 23. Available water-holding capacity

AWH Capacity	Class	Code
< 7.5 cm	Very low	VL
7.5 to < 15 cm	Low	L
15 to < 22.5 cm	Moderate	M
≥ 22.5 cm	High	H

4.5 Depth of Prolonged Water Saturation

Task on the scoresheet: Determine the depth of reductimorphic features comprising $\geq 10\%$ of the exposed area.

Critical for understanding the effects of flooding, partitioning of water, drainage, habitat, water purification, and construction. Landscape position, slope gradient, infiltration rate, surface runoff, saturated hydraulic conductivity, porosity, and permeability are significant factors influencing the long-term average depth up to which the soil profile is water-saturated in average precipitation years. Saturation may be permanent or temporary. Prolonged periods of saturation cause the absence of oxygen, and microorganisms reduce Mn and Fe, producing reductimorphic features. Reductimorphic

features that comprise $\geq 10\%$ of the exposed area are considered indicators of prolonged water saturation in this handbook. Use *Table 24*.

Table 24. Shallowest depth where evidence of prolonged water saturation occurs

Criteria	Depth class	Code
Reductimorphic features comprising $\geq 10\%$ of the exposed area occur in some part ≤ 25 cm	Very shallow	VS
Reductimorphic features comprising $\geq 10\%$ of the exposed area occur in some part between > 25 and ≤ 50 cm but not shallower	Shallow	S
Reductimorphic features comprising $\geq 10\%$ of the exposed area occur in some part between > 50 and ≤ 100 cm but not shallower	Moderately deep	MD
Reductimorphic features comprising $\geq 10\%$ of the exposed area occur in some part between > 100 and ≤ 150 cm but not shallower	Deep	D
Reductimorphic features comprising $\geq 10\%$ of the exposed area do not occur at ≤ 150 cm (see Notes below)	Very deep	VD

Notes: If reductimorphic features comprising $\geq 10\%$ of the exposed area do not occur within the specified depth for judging and that depth is < 150 cm, then assume a class of very deep. Short duration free water in the soil pit may have been taken out by the organizers to allow participants to easily enter the pit to describe the profile. Therefore, the current depth of a water table does not necessarily indicate the depth where prolonged saturation occurs.

4.6 Surface Runoff

Task on the scoresheet: Determine the surface runoff class according to *Table 25* based on slope and the hydraulic conductivity class of the *mineral surface layer* (*Section 4.3*).

Soils with low amounts of bare soil exposed have slower runoff rates, and those with structureless massive or crusted surfaces have higher runoff. Adjust to one slower class if there is $< 10\%$ bare ground (on average) between the slope stakes, unless the class is already *Very slow* or *Ponded*. Adjust to one more rapid class if the soil structure type in the mineral surface layer is structureless and massive or the surface is crusted in most of the area between the slope stakes, unless the rate is already *Ponded*. Soils with Low hydraulic conductivity class near the mineral soil surface have higher runoff during prolonged heavy rainfall events once all the pores become filled. Adjust to one more rapid class if the limiting (slowest) hydraulic conductivity class within 50 cm of mineral soil surface is Low.

Table 25. Surface runoff classes (Southeast Regional Collegiate Soil Contest Official Handbook, modified)

Slope	Hydraulic conductivity class of the <i>mineral surface layer</i>		
	High (H)	Moderate (M)	Low (L)
Depression	Ponded (P)	Ponded (P)	Ponded (P)
0 to 1% slope	Very slow (VS)	Very slow (VS)	Slow (S)
> 1 to 2% slope	Very slow (VS)	Slow (S)	Medium (M)
> 2 to 6% slope	Slow (S)	Medium (M)	Rapid (R)
> 6 to 12% slope	Medium (M)	Rapid (R)	Very rapid (VR)
$> 12\%$ slope	Rapid (R)	Very rapid (VR)	Very rapid (VR)

4.7 Potential Water Erosion

Task on the scoresheet: Determine the potential water erosion class according to *Table 26*. Take the surface runoff class (*Table 25*) and move across the table to the column with the texture of the mineral surface layer.

The potential for losses by water erosion is influenced mainly by the texture of the mineral surface layer and the amount of surface runoff at a site, which already takes surface cover, slope, structure of the mineral surface layer, and near-surface conductivity classes into account.

Table 26. Classes of potential water erosion (Southeast Regional Collegiate Soil Contest Official Handbook, modified)

Surface runoff class	Mineral surface layer texture			
	S, LS and layers with $\geq 12\% C_{org}$	SCL, SC	SL, CL, C, SiC	L, Si, SiL, SiCL
Ponded (P)	Very Low (VL)	Very Low (VL)	Very Low (VL)	Very Low (VL)
Very slow (VS)	Very Low (VL)	Very Low (VL)	Low (L)	Low (L)
Slow (S)	Very Low (VL)	Low (L)	Low (L)	Medium (M)
Medium (M)	Very Low (VL)	Low (L)	Medium (M)	High (H)
Rapid (R)	Low (L)	Medium (M)	High (H)	Very High (VH)
Very Rapid (VR)	Medium (M)	High (H)	Very High (VH)	Very High (VH)

5 INTERPRETATIONS

The following soil interpretation tables are used to allocate soils to a category of suitability for specific land uses. Note that N, P, K fertility and pH are not assessed. Other data will be provided on pitcards at each pit if relevant. Some factors below are correctable by drainage to lower depth to prolonged saturation or terracing to reduce slope angle. Soluble salt content may be managed with proper irrigation with less saline water sources. The occurrence of flooding from streams or ponding makes the land unsuitable for all of the listed uses, so we will assume each soil pit site is protected from flooding. Depth values are related to the mineral soil surface (see *Section 1.7*), unless stated otherwise.

For most rapid analysis of limitations in any table below:

1. Start in the right-hand Class 3 column of the tables.
2. Read down the Class 3 column, checking the criteria.
3. If one or more factor criteria in the Class 3 column is met, list the appropriate factor number(s) in the Class 3 box on the scoresheet. Do not make any marks in the Class 2 or Class 1 box.
4. If no factor criteria in the Class 3 column are met, check the Class 2 column. If one or more factor criteria is met in the Class 2 column, list the appropriate factor number(s) in the Class 2 box on the scoresheet. Do not make any marks in the Class 3 or Class 1 box.
5. If no factor criteria in the Class 3 or Class 2 column are met, place a checkmark in the Class 1 box on the scoresheet.

Task on the scoresheet: Use the respective *Table* to assess the suitability for the respective land use, with the most limiting factor of any of the soil and land characteristics to allocate a soil to a suitability class.

5.1 Potato Production

Table 27. Suitability factors for potato production

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Texture class or organic layers with the highest combined thickness* in the upper 20 cm; rock fragments in any layer within the upper 20 cm	Organic layers, SL, SiL, or L	LS, Si, CL, SiCL or SCL	S, SC, SiC or C; $\geq 15\%$ rock fragments
2. Effective soil depth class	VD or D	MD	S or VS
3. Available water-holding capacity class	H or M	L	VL
4. Depth of prolonged water saturation	VD or D	MD or S	VS
5. Potential water erosion	<u>P, VL</u>	L	M, H, VH

* To evaluate, sum the thickness of all listed textures or organic layers within each suitability class, and then select the highest amount. In this section, *organic layers* are defined by having $\geq 12\%$ soil organic carbon.

5.2 Paddy Rice Production

Table 28. Suitability factors for paddy rice production

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Electrical conductivity (dS/m, saturation extract) in any layer within the upper 20 cm	≤ 2	> 2 to 4	> 4
2. Texture class or organic layers with the highest combined thickness* in the upper 20 cm; rock fragments in any layer within the upper 20 cm	Organic layers and all other textures	SL	S or LS; $\geq 15\%$ rock fragments
3. Effective soil depth class	VD or D	MD or S	VS
4. Depth (cm) to shallowest layer with low hydraulic conductivity class	≤ 25	> 25 to 50	> 50
5. Slope gradient (%)	≤ 1	> 1 to 2	> 2

Note: It is assumed that controlled flooding/irrigation is practiced on the field in most years.

Data on *electrical conductivity* will be provided on pitcards of each pit if relevant.

* To evaluate, sum the thickness of all listed textures or organic layers within each suitability class, and then select the highest amount. In this section, *organic layers* are defined by having $\geq 12\%$ soil organic carbon.

5.3 Wheat Production

Table 29. Suitability factors for wheat production

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Electrical conductivity (dS/m, saturation extract) in any layer within the upper 20 cm	≤ 4	>4 to 8	> 8
2. Texture class or organic layers with the highest combined thickness* in the upper 20 cm; rock fragments in any layer within the upper 20 cm	SL, L, SiL, SCL, or CL	LS, SC, Si, or SiCL	Organic layers, S. C or SiC; ≥ 15% rock fragments
3. Available water-holding capacity class	H or M	L	VL
4. Depth of prolonged water saturation	VD or D	MD or S	VS
5. Potential water erosion	P, VL, or L	M	H or VH

Note: Data on *electrical conductivity* will be provided on pitcards of each pit if relevant.

* To evaluate, sum the thickness of all listed textures or organic layers within each suitability class, and then select the highest amount. In this section, *organic layers* are defined by having ≥ 12% soil organic carbon.

5.4 Corn (Maize) Production

Table 30. Suitability factors for corn (maize) production

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Electrical conductivity (dS/m, saturation extract) in any layer within the upper 20 cm	≤ 2	> 2 to 4	> 4
2. Texture class or organic layers with the highest combined thickness* in the upper 20 cm; rock fragments in any layer within the upper 20 cm	Organic layers, SL, L, SCL, or CL	LS, SC, Si, SiL, or SiCL	S. C or SiC; ≥ 15% rock fragments
3. Available water-holding capacity class	H	M	L or VL
2. Depth of prolonged water saturation	VD or D	MD or S	VS
5. Potential water erosion	P or VL	L	M, H, or VH

Note: Data on *electrical conductivity* will be provided on pitcards of each pit if relevant.

* To evaluate, sum the thickness of all listed textures or organic layers within each suitability class, and then select the highest amount. In this section, *organic layers* are defined by having ≥ 12% soil organic carbon.

5.5 Soybean Production

Table 31. Suitability factors for soybean production

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Electrical conductivity (dS/m, saturation extract) in any layer within the upper 20 cm	≤ 4	> 4 to 8	> 8
2. Texture class or organic layers with the highest combined thickness* in the upper 20 cm; rock fragments in any layer within the upper 20 cm	SL, L, SiL, SCL, or CL	LS, SC, Si, or SiCL	Organic layers, S. C or SiC; ≥ 15% rock fragments
3. Available water-holding capacity class	H	M	L or VL
4. Depth of prolonged water saturation	VD or D	MD	S or VS
5. Potential water erosion	P or VL	L	M, H, or VH

Note: Data on *electrical conductivity* will be provided on pitcards of each pit if relevant.

* To evaluate, sum the thickness of all listed textures or organic layers within each suitability class, and then select the highest amount. In this section, *organic layers* are defined by having ≥ 12% soil organic carbon.

5.6 Home Gardens

Table 32. Suitability factors for home gardens

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Electrical conductivity (dS/m, saturation extract) in any layer within the upper 20 cm	≤ 1	> 1 to 2	> 2
2. Texture class or organic layers with the highest combined thickness* in the upper 20 cm; rock fragments in any layer within the upper 20 cm	Organic layers, SL, L, SCL, or CL	LS, SC, Si, SiL, or SiCL	S. C or SiC; ≥ 15% rock fragments
3. Available water-holding capacity class	H	M or L	VL
4. Depth of prolonged water saturation	VD or D	MD or S	VS
5. Potential water erosion	P or VL	L	M, H, or VH

Note: Data on *electrical conductivity* will be provided on pitcards of each pit if relevant.

* To evaluate, sum the thickness of all listed textures or organic layers within each suitability class, and then select the highest amount. In this section, *organic layers* are defined by having ≥ 12% soil organic carbon.

5.7 On-Site Septic Absorption Fields

Table 33. Suitability factors for on-site septic absorption fields

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Weighted average rock fragment percentage 0 to 100 cm	≤ 15	> 15 to 60	> 60
2. Depth (cm) to bedrock (ST: R or Cr, WRB: R) or any cemented layer (m)	> 150	> 100 to 150	≤ 100
3. Depth (cm) to the layer with <i>Low</i> hydraulic conductivity	> 150	≤ 50 or > 100 to ≤ 150	> 50 to ≤ 100
4. Depth of prolonged water saturation	VD	D	MD, S or VS
5. Slope gradient (%)	≤ 6	> 6 to 12	> 12

5.8 Buildings without Basements

Table 34. Suitability factors for buildings without basements

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Depth to organic layers, ≥ 10 cm thick	> 150	> 100 to 150	≤ 100 cm
2. Shrink-swell potential in summer (surface cracks; slickensides anywhere in the upper 100 cm)	No surface cracks and no slickensides	Surface cracks ≤ 2 cm wide present but no slickensides	Surface cracks > 2 cm wide and/or slickensides present
3. Depth to hard rock/bedrock (R) or any cemented layer (m)	> 150	> 50 to 150	≤ 50
4. Depth of prolonged water saturation	VD	D or MD	S or VS
5. Slope gradient (%)	≤ 6	> 6 to 20	> 20

Note: Data on *surface cracks* will be provided on pitcards of each pit if relevant.

5.9 Local Roads for Community Planning

Table 35. Suitability factors for local roads for community planning

Factor	Suitability		
	Class 1 Optimal	Class 2 Suitable	Class 3 Unsuitable
1. Susceptibility to frost action in non-tropical regions (texture anywhere within the upper 100 cm)	S, LS or SL	All other textures	Si, SiL or SiCL
2. Shrink-swell potential in summer (surface cracks; slickensides anywhere in the upper 100 cm)	No surface cracks and no slickensides	Surface cracks $\leq$ 2 cm wide present but no slickensides	Surface cracks $>$ 2 cm wide and/or slickensides present
3. Depth to hard rock/bedrock (R) or any cemented layer (m)	> 150	> 100 to 150	≤ 100
4. Depth of prolonged water saturation	VD, D or MD	S	VS
5. Slope gradient (%)	≤ 6	> 6 to 20	> 20

Note: Data on *surface cracks* will be provided on pitcards of each pit if relevant.

6 DIAGNOSTICS AND SOIL CLASSIFICATION

The competitors must classify the soils according to both Soil Taxonomy (ST, 13th edition, 2022) and World Reference Base for Soil Resources (WRB, 4th edition, 2022). Additional data necessary for the classification will be provided on the pitcard at each pit. Please see the **Simplified WRB Diagnostics addendum** and **Keys to Soil Taxonomy addendum** for information specific to this contest.

Note to Contest Host:

For Soil Taxonomy, a Keys to Soil Taxonomy addendum will be supplied. For WRB, a Simplified WRB Diagnostics addendum will be supplied. The scoresheet should be adjusted based on the practice and contest soils to be seen, plus any other commonly found soils in the region, even if they are not shown.

For practice sites only, a national system can be given and explained. This would be of great interest to the students and coaches.

6.1 World Reference Base for Soil Resources (2022)

6.1.1 Diagnostic horizons/properties/materials

Task on the scoresheet: Use a cross (X) to select all the diagnostic horizons/properties/materials that apply to the profile within the specified judging depth. For detailed information on the horizons/properties/materials see Chapter 3 of the WRB.

6.1.2 Reference Soil Group

Task on the scoresheet: Use a cross (X) to select the Reference Soil Group (RSG). Use the key to the RSGs in Chapter 4 of the WRB.

6.1.3 Qualifiers

Task on the scoresheet: Use a cross (X) to select all the qualifiers that apply to the profile within the specified judging depth. For detailed information on qualifiers see Chapter 5 of the WRB.

6.2 Soil Taxonomy (2022)

6.2.1 Epipedon

Task on the scoresheet: Use a cross (X) to select the epipedon that applies to the profile. For detailed information on the epipedon definitions see pages 7-19 of the Keys to Soil Taxonomy.

6.2.2 Subsurface horizons

Task on the scoresheet: Use a cross (X) to select all the subsurface horizons that apply to the profile within the specified judging depth. For detailed information on the subsurface horizon definitions see pages 19-31 of the Keys to Soil Taxonomy.

6.2.3 Diagnostic characteristics

Task on the scoresheet: Use a cross (X) to select all the diagnostic characteristics that apply to the profile within the specified judging depth. For detailed information on the diagnostic characteristic definitions see pages 31-55 of the Keys to Soil Taxonomy.

6.2.4 Order

Task on the scoresheet: Use a cross (X) to select the soil order. For the Keys to Soil Orders see pages 65-70 of the Keys to Soil Taxonomy.

6.2.5 Suborder

Task on the scoresheet: Use a cross (X) to select the soil suborder.

6.2.6 Great Group

Task on the scoresheet: Use a cross (X) to select the soil great group.

7 REFERENCES

IUSS Working Group WRB (2022): World Reference Base for Soil Resources. International soil classification system for naming soils and creating legends for soil maps. 4th edition. International Union of Soil Sciences (IUSS), Vienna, Austria.

Soil Survey Staff (2024): Field Book for describing and sampling soils. Version 4.0. USDA Natural Resources Conservation Service.

Soil Survey Staff (2022): Keys to Soil Taxonomy. 13th edition. USDA Natural Resources Conservation Service.

8 SCORING INFORMATION

1. SITE DESCRIPTION		
Land use	Only one answer for each is correct. For the correct answer, 1 point will be awarded. For an incorrect answer, 0 points will be awarded.	
Landform		
Slope position		
Slope gradient	If the gradient is $\leq 5\%$, 1 point will be awarded if the answer is $\pm 1\%$ correct compared to the value estimated by the organizers. If the gradient is $> 5\%$, 1 point will be awarded if the answer is $\pm 2\%$ correct. If the area is flat, only 0 is the correct answer.	
Rock outcrops	1 point will be awarded if the answer is $\pm 10\%$ correct compared to the values estimated by the organizers. If rock outcrops or surface rock fragments are absent, only 0 is the correct answer.	
Surface rock fragments		
Human-made surface unevenness	Only one answer is correct. For the correct answer, 1 point will be awarded. For an incorrect answer, 0 points will be awarded.	
Litter layer: area covered	1 point will be awarded if the answer is $\pm 10\%$ correct compared to the value estimated by the organizers. If a litter layer is absent, only 0 is the correct answer.	
Litter layer: average thickness	If the litter layer is ≤ 5 cm thick, 1 point will be awarded if the answer is ± 1 cm correct compared to the value estimated by the organizers. If the litter layer is > 5 cm thick, 1 point will be awarded if the answer is ± 2 cm correct. If a litter layer is absent, only 0 is the correct answer.	
2. SOIL DESCRIPTION		
Layer lower depth	The threshold of correct lower depth readings depends on the distinctness of the boundary. Abrupt: ± 1 cm Clear: ± 3 cm Gradual/Diffuse: ± 8 cm. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.	
Layer boundary distinctness	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.	
Layer boundary shape (topography)	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.	
Texture class	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.	
Texture clay %	A scaled range for correct answers compared to values obtained from laboratory data will be used according to:	
	Measured clay content	Range for grading
	< 20 %	± 4 %
	20 to 40 %	± 6 %
	> 40 %	± 8 %
Texture sand %	A scaled range for correct answers compared to values obtained from laboratory data will be used according to:	
	Measured sand content	Range for grading
	< 20 %	± 6 %
	20 to 40 %	± 8 %
	> 40 %	± 10 %

Rock fragments and artefacts: %	A scaled range for correct answers compared to the value estimated by the organizers will be used. 2 points will be awarded if the answer is $\pm 5\%$ correct, 1 point if the answer is $\pm 10\%$ correct. If rock fragments are absent, only 0 is the correct answer.
Rock fragments and artefacts: weighted average %	The points are awarded if the calculations based on the rock fragment contents and the layer thicknesses estimated by the contestants are correct. A scaled range will be used. 2 points will be awarded if the answer is $\pm 5\%$ correct, 1 point if the answer is $\pm 10\%$ correct. If rock fragments are absent, only 0 is the correct answer.
Structure type	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Structure grade	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Soil color hue	Only one answer is correct. For the correct answer, 1 point will be awarded. For an incorrect answer, 0 points will be awarded.
Soil color value	For the correct answer, 2 points will be awarded. If the answer is lower or higher than the correct one by 1 value category, 1 point will be awarded. If the answer is lower or higher than the correct one by 2 value categories, 0 points will be awarded.
Soil color chroma	For the correct answer, 2 points will be awarded. If the answer is lower or higher than the correct one by 1 chroma category, 1 point will be awarded. If the answer is lower or higher than the correct one by 2 chroma categories, 0 points will be awarded.
Carbonate content	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Secondary carbonates: type	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Secondary carbonates: %	1 point will be awarded if the answer is $\pm 5\%$ correct, compared to the value estimated by the organizers. If secondary carbonates are absent, only 0 is the correct answer.
Redox features: location of oximorphic features	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Redox features: location of reductimorphic features	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Redox features: OOA + OOE: %	1 point will be awarded if the answer is $\pm 10\%$ correct, compared to the value estimated by the organizers. If oximorphic features are absent, only 0 is the correct answer.
Redox features: oximorphic total: %	1 point will be awarded if the answer is $\pm 10\%$ correct, compared to the value estimated by the organizers. If oximorphic features are absent, only 0 is the correct answer.
Redox features: reductimorphic total: %	1 point will be awarded if the answer is $\pm 5\%$ correct, compared to the value estimated by the organizers. If reductimorphic features are absent, only 0 is the correct answer.
Aggregate surface features: type	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.
Aggregate surface features: %	1 point will be awarded if the answer is $\pm 5\%$ correct, compared to the value estimated by the organizers.
Parent material	Only one answer is correct. For the correct answer, 2 points will be awarded. For an incorrect answer, 0 points will be awarded.

3. LAYER DESIGNATIONS	
Stratum	Only one answer is correct. For the correct answer, 1 point will be awarded. For an incorrect answer, 0 points will be awarded.
Master symbol	The correct answer is either one or two master symbols and any symbol used for combination horizons (such as E&B, B/E). They are awarded two points if every symbol they record is correct. They lose 1 point for any symbol that is incorrect or missing. If no symbols are correct, they receive zero points. Examples: The correct answer is A, and they put AB, they receive 1 point. The correct answer is B/E and they put B, they receive 0 points.
Suffix(es)	If the correct answer is 3 or less suffixes, 3 points will be awarded for a fully correct answer. 1 point is deducted for any missing or wrong suffix. However, if the correct answer has 4 or more suffixes then 1 point is awarded per correct suffix and 1 point is deducted per wrong answer (missing answers are not deducted, and the maximum score is 3). The sequence of the suffixes will not be considered in grading.
Numeral	Only one answer is correct. For the correct answer, 1 point will be awarded. For an incorrect answer, 0 points will be awarded.
4. SOIL PROFILE CHARACTERISTICS	
Effective soil depth	Only one answer is correct. For the correct answer, 3 points will be awarded. For an incorrect answer, 0 points will be awarded.
Type of restrictive layer	
Hydraulic conductivity: Surface layer	
Hydraulic conductivity: Layer with lowest cond.	
AWC	
Depth of prolonged water saturation	
Surface runoff	
Potential water erosion	
5. INTERPRETATIONS	
All	For every interpretation, only one answer is correct. For the correct answer, 3 points will be awarded. For an incorrect answer, 0 points will be awarded.
6.1. DIAGNOSTICS AND SOIL CLASSIFICATION - WRB	
Diagnostic horizons	Multiple answers might be correct. If correct horizons are marked, 10 points will be awarded for each one. If incorrect horizons are marked, -10 points will be awarded. The overall score cannot be lower than 0.
Diagnostic properties	Multiple answers might be correct. If correct properties are marked, 5 points will be awarded for each one. If incorrect properties are marked, -5 points will be awarded. The overall score cannot be lower than 0.
Diagnostic materials	Multiple answers might be correct. If correct materials are marked, 5 points will be awarded for each one. If incorrect materials are marked, -5 points will be awarded. The overall score cannot be lower than 0.
RSG	Only one answer is correct. For the correct RSG, 15 points will be awarded. For an incorrect RSG, 0 points will be awarded.

Qualifiers	Multiple answers might be correct. If correct qualifiers are marked, 5 points will be awarded for each one. If incorrect qualifiers are marked, -5 points will be awarded. The overall score cannot be lower than 0.
6.2 DIAGNOSTICS AND SOIL CLASSIFICATION - SOIL TAXONOMY	
Epipedon	Only one answer is correct. For the correct epipedon, 10 points will be awarded. For an incorrect epipedon, 0 points will be awarded.
Subsurface horizons	Multiple answers might be correct. If correct horizons are marked, 5 points will be awarded for each one. If incorrect horizons are marked, -5 points will be awarded. The overall score cannot be lower than 0.
Diagnostic characteristics	Multiple answers might be correct. If correct characteristics are marked, 5 points will be awarded for each one. If incorrect characteristics are marked, -5 points will be awarded. The overall score cannot be lower than 0.
Order	Only one answer is correct. For the correct Order, 10 points will be awarded. For an incorrect Order, 0 points will be awarded.
Suborder	Only one answer is correct. For the correct Suborder, 5 points will be awarded. For an incorrect Suborder, 0 points will be awarded.
Great Group	Only one answer is correct. For the correct Great Group, 5 points will be awarded. For an incorrect Great Group, 0 points will be awarded.



SCORESHEET

Profile ID..... Contestant ID..... Describe.....horizons to a depth ofcm Nail at.....cm at the bottom of horizon..... (Additional nail at.....cm at the bottom of horizon.....)

1. SITE DESCRIPTION

SCORE:

Land Use	Landform	Slope	Rock outcrops	Surface rock fragments	Surface unevenness	Litter layer
Type (1)	Type (1)	Position (1)	% (1)	% (1)	Type (1)	Average thickn. (cm) (1)

2. PROFILE DESCRIPTION

SCORE:

Layer boundaries				Texture			Rock fragments		Structure		Color			Carbonates				Redoximorphic features					Aggregate surfaces		Par. mat.
No.	Lower Depth (cm) (2)	Distinct ness (2)	Shape (2)	Class (2)	Clay % (2)	Sand % (2)	% (2)	Weight. Aver. % (2)	Type (2)	Grade (2)	Hue (1)	Value (2)	Chroma (2)	Content (2)	Sec. carb. Type (2)	Sec. carb. % (1)	Oxim. Loc. (2)	Reducim. Loc. (2)	OOA + OOE % (1)	Oxim. total % (1)	Reducim. total % (1)	Type (2)	Type (2)		
1																									
2																									
3																									
4																									
5																									
6																									

SCORE:

3.1 LAYER DESIGNATIONS (WRB)

No.	Stratum (1)	Master (2)	Suffix(es) (3)	Numeral (1)
1				
2				
3				
4				
5				
6				

SCORE:

3.2 LAYER DESIGNATIONS (ST)

No.	Stratum (1)	Master (2)	Suffix(es) (3)	Numeral (1)
1				
2				
3				
4				
5				
6				

4. SOIL CHARACTERISTICS

SCORE:

Type of restrictive layer (3)	Effective soil depth (3)	Hydraulic conductivity Surface layer (3)	Hydraulic conductivity Layer with lowest cond. (3)	AWC (3)	Depth of prolonged water saturation (3)	Surface runoff (3)	Potential water erosion (3)
Bedrock	VS	High	High	VL	VS	P	VL
Cemented	S	Moderate	Moderate	L	S	VS	L
Densic	MD	Low	Low	M	MD	S	M
Massive clay	D			H	D	M	H
None	VD				VD	R	VH

10 CHARTS FOR ESTIMATING PERCENTAGES

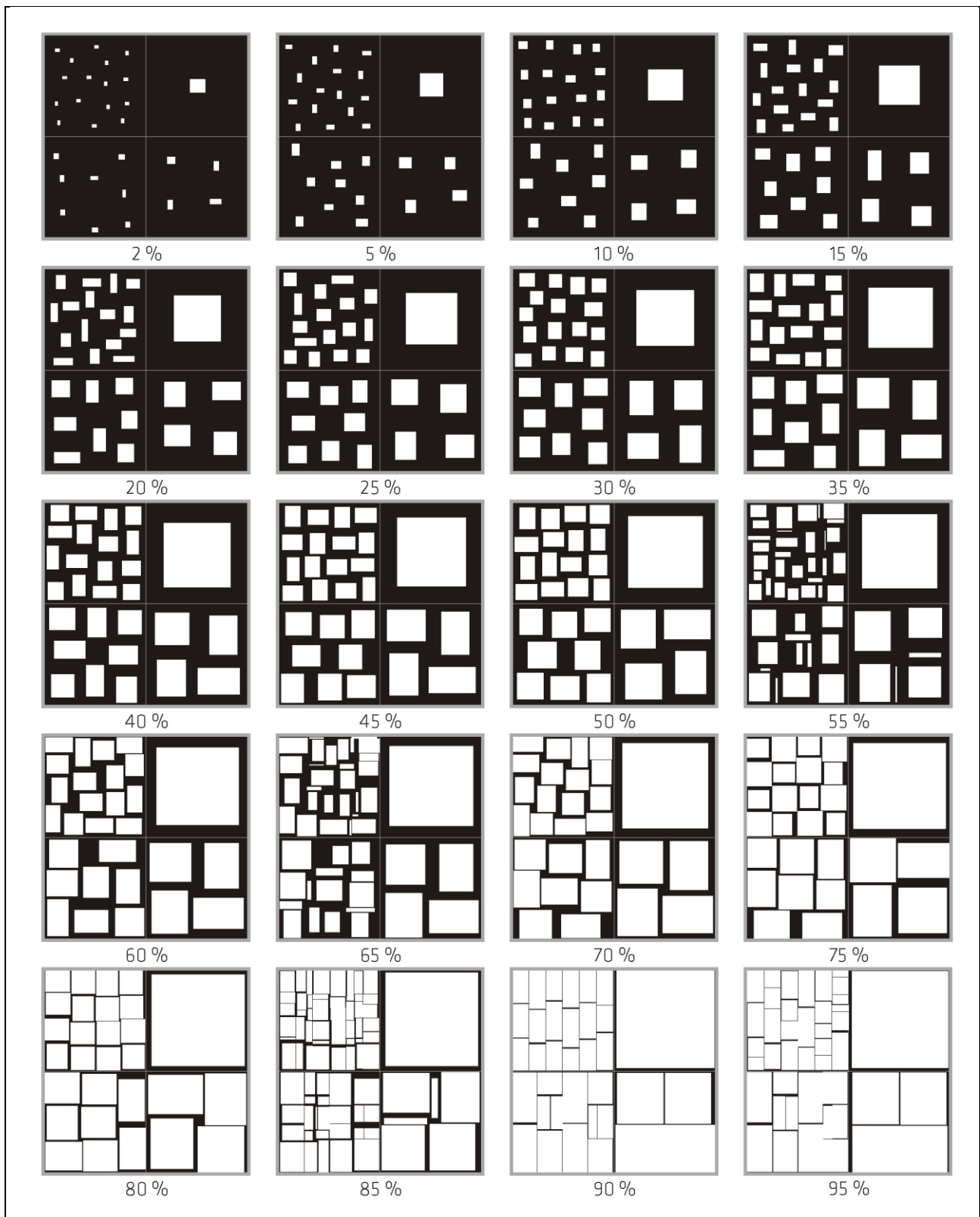


Figure 11. Charts for estimating percentages (IUSS Working Group WRB, Figure 8.12, modified)